

Institutions as Distributed Computations: A Unified Framework Integrating Extended Phenotype Theory, Evolutionary Game Theory, and Computational Complexity

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Abstract

Why do some legal institutions resist reform for decades while others adapt within a single legislative cycle? Existing accounts draw on veto player theory, path dependence, and interest group coalitions. None supplies a mechanism that simultaneously connects individual cognition, collective behavior, and formally specified structural constraints on feasible reform trajectories. This paper proposes such a mechanism through the integration of three research traditions: Extended Phenotype Theory (EPT), Evolutionary Game Theory (EGT), and computational complexity theory.

The integration is motivated by a convergence finding: Hamilton et al. (2026), working independently from the EPT-EGT research programme, argue that human societies are distributed computational systems whose organizational forms are shaped by the resource costs of collective problem-solving. Their framework, developed at the intersection of theoretical computer science and social science, maps social coordination phenomena onto formal problems from Boolean circuit theory, distributed computing, approximation complexity, and Byzantine Fault Tolerance. This paper argues that the Constitutional Lock-in Index (CLI), Institutional Hysteresis Rate (IHR), and Institutional Evolvability Index (IEI), developed within the EPT-EGT programme, are operationalizations of the circuit-theoretic concepts Hamilton et al. identify as central to institutional analysis.

The paper develops four main contributions. First, a formal mapping between CLI/IHR/IEI values and circuit-theoretic parameters (circuit depth, fault-propagation cost, non-uniform reconfiguration capacity) is proposed and its empirical implications derived. Second, Heteronomous Bayesian Updating (HBU) is reinterpreted as an advice-string mechanism in the sense of computational complexity theory, with specific implications for the timing of reform windows. Third, Zombie Law Equilibria are formally characterized as distributed systems that have exceeded their Byzantine Fault Tolerance threshold, which entails that they cannot be resolved by any internally generated deterministic protocol. Fourth, a research agenda for empirical testing of the synthesis is specified across three methodological platforms: the existing sixty-case international dataset, a proposed natural-experiment design exploiting disruptions to legal professional socialization, and a planned extension of the 816-agent computational model.

The paper argues that the EPT-EGT-computation synthesis is more explanatorily powerful than any of its three components separately, because each component supplies what the others lack: EPT supplies the replication mechanism, EGT supplies the selection dynamics, and computational complexity supplies formal constraints on feasible trajectories that neither evolutionary framework can specify on its own.

Keywords: Extended Phenotype Theory, Evolutionary Game Theory, Constitutional Lock-in Index, computational complexity, Byzantine Fault Tolerance, Heteronomous Bayesian Updating, institutional evolution, distributed computation, circuit depth

JEL codes: B52, D72, D74, K10, O17, P16

1. Introduction

The puzzle is precise. In September 2017, the Brazilian Congress passed a comprehensive reform of the *Consolidacao das Leis do Trabalho* (CLT) in a single legislative session. The reform altered twenty-six years of normative sediment accumulated since the 1988 Constitution, recalibrated the relationship between collective bargaining agreements and statutory minimums, and modified the procedural rules governing labor litigation. It was implemented without constitutional challenge reaching the Supreme Federal Tribunal on the core provisions. The reform was contested, but it held.

Argentina has been attempting a structurally comparable reform since 1991. Not one reform. Twenty-three separate attempts, spanning seven governments from Menem's convertibility program through the current administration, across ideological positions from orthodox liberalism to kirchnerismo to libertarianism. The provisions at stake, concentrated in Article 14bis of the 1853-1994 National Constitution, have survived every attempt without substantive modification. The single partial success, the flexibilization decrees of 1996-1998, was rolled back by a combination of judicial injunctions and subsequent executive action within thirty-six months.

Chile reformed its labor regime in 1990, 2001, and 2015. Three successful reforms in twenty-five years, each constructing on the previous without triggering systemic institutional resistance. Spain reformed its constitutional labor framework in 1994 and again in 2012 under structurally different political conditions; both reforms held.

The standard explanatory apparatus cannot account for this pattern. Veto player theory (Tsebelis, 2002) predicts that Argentina, with a bicameral legislature and formal supermajority requirements for constitutional amendment, should be harder to reform than Brazil under its 1988 Constitution. In practice, Brazil reformed successfully and Argentina did not. Union density is comparable across Argentina and Brazil; if union power were the operative variable, the outcomes should not diverge so sharply. Path dependence (North, 1990; Pierson, 2004) describes persistence but does not generate the specific predictions necessary to explain why persistence obtains in some cases and not others. Interest group coalition theory predicts resistance where coalitions are strong; it cannot explain the geometry of failure, why certain reform attempts get further than others, why certain institutional layers are reliably activated while others are not.

This paper argues that the divergence in reform outcomes follows a pattern predicted by computational complexity theory, and that this prediction was not derivable from the evolutionary frameworks developed in prior work within this research programme. The argument proceeds in two stages. In the first stage, I establish that the EPT-EGT research programme and the computational social science tradition represented by Hamilton et al. (2026) have converged independently on the same structural observations about institutional persistence. In the second stage, I develop the formal integration and use it to generate predictions that extend beyond what either tradition alone could produce.

A terminological note before proceeding. The claim that institutions are "computational" is not a metaphor in this paper. The claim is formal: the resource requirements for institutional reform satisfy the mathematical structure of the resource requirements for distributed consensus under Byzantine fault conditions, and this structural identity generates falsifiable empirical

predictions. Where I use computational vocabulary (circuit depth, advice strings, Byzantine faults), I intend the formal mathematical referent, not an analogy.

2. Theoretical Background and Epistemological Positioning

2.1 The EPT-EGT Research Programme: Hard Core and Protective Belt

This paper belongs to a research programme in the sense of Lakatos (1978). The hard core of the programme holds that legal norms are cultural replicators whose persistence, drift, and extinction follow evolutionary dynamics governed by Extended Phenotype Theory (Dawkins, 1982) and Evolutionary Game Theory (Maynard Smith and Price, 1973; Maynard Smith, 1982). The protective belt consists of a set of specific metrics and formal constructs that are individually revisable without abandoning the core evolutionary commitment.

The metrics constituting the current protective belt include: the Constitutional Lock-in Index (CLI), a composite measure of the density of institutional infrastructure defending specific normative provisions; the Institutional Hysteresis Rate (IHR), a measure of the rate at which institutional deformation persists and propagates irreversibly after a perturbation; the Institutional Evolvability Index (IEI), a four-component measure of a system's capacity for adaptive reconfiguration under environmental pressure; Heteronomous Bayesian Updating (HBU), a formal account of normative prior formation through authority-reaction observation rather than behavioral frequency; the Asymmetric Intentionality Theory (AIT) and its generalization (GIMT), formal accounts of intentionality mismatch between normative producers and normative targets; and the Dennett-Nash Gap, a measure of the divergence between cooperative optima and Nash equilibria in norm-governed populations.

Popper (1959) supplies the demarcation criterion: the hypotheses formulated within this programme are falsifiable and their empirical tests are specified. The programme has generated falsifiable predictions validated against empirical data in prior publications. CLI scores above 0.70 predicted reform failure at a 95% rate across the sixty-case international dataset (Lerer, 2025a). The HBU mechanism predicted that professional socialization disruptions should precede reform windows; this prediction has not yet been systematically tested but is specified in Section 5.2 below.

Hull (1988) and Campbell (1974) provide the epistemological grounding for applying selection-based reasoning to cultural and institutional entities. The position of this programme with respect to Kuhn (1962) is that legal theory is currently in a state of paradigm crisis (Lerer, 2025e): the accumulation of anomalies that positivist and natural law frameworks cannot explain without auxiliary hypothesis multiplication has reached the threshold where revolutionary science becomes possible. The present paper does not argue this position; it proceeds under it.

Epistemological positioning. This paper treats the convergence between the EPT-EGT programme and the computational social science tradition of Hamilton et al. (2026) as evidence of validity that neither tradition can generate alone. When two research traditions developed under different methodological constraints, drawing on different primary literatures, and asking different research questions arrive at structurally similar claims about the same phenomena, the convergence is itself empirical evidence that the claims track something real. The falsifiability criterion (Popper, 1959) applies to each specific claim derived from the synthesis, and Section 5 specifies the tests.

2.2 The Computational Turn in Social Science

The application of information-theoretic and computational concepts to social systems has a long history. Hayek (1945) characterized the market as an information-processing system, though without formal specification. Simon (1962) analyzed the architecture of complexity in terms of hierarchical modular structure, anticipating the circuit-theoretic framework of later computer science. Von Neumann's work on reliable computation from unreliable components (Von Neumann, 1956) laid the foundation for fault-tolerance theory that eventually became Byzantine Fault Tolerance.

The formal computational complexity tradition entered social science analysis more slowly. Papadimitriou (2007) examined the computational complexity of finding Nash equilibria, establishing that the problem is PPAD-complete. Daskalakis, Goldberg and Papadimitriou (2009) refined this result. The implication is that in general, rational agents cannot be expected to find Nash equilibria through decentralized computation: the problem is computationally intractable even for agents with unlimited reasoning capacity but no communication. This result bears directly on any theory of institutional equilibrium that postulates agents converging on Nash outcomes through deliberation.

Hebert-Dufresne et al. (2025) represent a recent extension of this tradition: they model governance as a complex satisfiability problem, implemented in a social hypergraph, and find that intermediately decentralized governance structures achieve the most efficient solutions. Their finding that "small overlapping decision groups making specific decisions and sharing information" (Hebert-Dufresne et al., 2025, p. 14) produce better outcomes than either fully centralized or fully distributed governance maps onto the NC complexity class: problems efficiently solvable in parallel by circuits of logarithmic depth.

Hamilton et al. (2026) extend this programme into a broader theoretical perspective. Their key contributions for the present analysis are: the identification of non-uniform computation as the formal analog of institutional restructuring with scale; the characterization of advice strings as the analog of cultural knowledge and norms; the analysis of Byzantine Fault Tolerance as a model for institutional robustness against defection; and the observation that average-case rather than worst-case complexity is the appropriate complexity paradigm for social systems facing typical rather than adversarial environments.

The gap in all of this prior work, which the present paper addresses, is the connection to empirically measured institutional structures. The computational social science tradition has developed formal frameworks but has not yet connected them to the specific metrics of the EPT-EGT programme. The EPT-EGT programme has developed and validated empirical metrics but has not yet derived formal resource-constraint predictions from them. The synthesis generates both.

2.3 The Evolutionary and Computational Traditions Compared

Table 1 summarizes the structural parallels between the two traditions that the present paper develops.

Dimension	EPT-EGT Programme	Computational Complexity
Persistence mechanism	Evolutionary stability (ESS); replicator fitness exceeds invasion threshold	Circuit depth exceeds NC threshold; computation requires superlogarithmic sequential steps
Prior formation	HBU: reaction-based Bayesian learning during professional socialization	Advice strings: precomputed size-dependent information reducing circuit depth for subsequent instances
Irreversibility	IHR: rate of irreversible institutional deformation after perturbation	Fault propagation cost; deficit of error-correcting redundancy relative to fault rate
Adaptability	IEI: four-component evolvability index (V, S, C, M)	Non-uniform computation: capacity to adopt different circuit architecture for different input sizes
Pathological equilibrium	Zombie Law Equilibrium: CLI > 0.85, fragmentation > 0.60	BFT threshold violation: faulty-node fraction exceeds (n-1)/3 under synchronous protocol
Reform failure	Fitness disadvantage of reform meme relative to incumbent	Computational intractability: reform requires exponential circuit depth or BFT-impossible consensus

Table 1: Structural correspondences between the EPT-EGT research programme and computational complexity theory.

3. Relevant Results from Computational Complexity Theory

3.1 Circuit Complexity: Size, Depth, and the NC Hierarchy

A Boolean circuit C over inputs $x_1, \dots, x_n$ is a directed acyclic graph in which each internal node represents a logic gate (AND, OR, NOT, or a generalization), each leaf represents an input bit or a constant, and each sink represents an output bit. The computation of C on an input string x is the assignment of Boolean values to all nodes according to the gate function, propagated from leaves to sinks.

Two complexity measures are central to the present analysis. The size $|C|$ of a circuit is the total number of gates. The depth $d(C)$ of a circuit is the length of the longest path from any input leaf to any output sink. Size measures the total computational work required; depth measures the minimum number of sequential steps, and hence the irreducible latency before the output can be determined regardless of how much parallel computation is available.

The class NC (Nick's Class) formalizes the problems efficiently solvable in parallel: NC consists of all problems solvable by circuits of polynomial size and polylogarithmic depth. More precisely, NC^i consists of problems solvable by polynomial-size circuits with depth $O(\log^i n)$. The class $NC = \bigcup_i NC^i$ characterizes problems that admit fast parallel solutions. Computing the majority of n bits, sorting n elements, and evaluating arithmetic expressions all belong to NC.

Problems outside NC require either circuits of superpolynomial size or circuits of superpolylogarithmic depth. In the latter case, the computation has irreducible sequential bottlenecks: no matter how many processors are available, the computation cannot complete in polylogarithmic time because there are chains of data dependencies that must be resolved sequentially. The

formal connection to institutional analysis is the thesis of Section 4.1: institutional reform processes with high CLI correspond to computations with superpoly-logarithmic depth.

3.2 Non-Uniform Computation and Advice Strings

Uniform computation, exemplified by the Turing Machine, applies a single computational procedure to inputs of all sizes. Non-uniform computation allows the procedure to vary with input size; the most important formal model is the Boolean circuit family, in which a separate circuit C_n is provided for inputs of length n . This distinction captures an important feature of institutional systems: the computational procedure appropriate for a specific reform depends on the type and scale of the reform, not just its content.

Advice complexity (Arora and Barak, 2009) formalizes the role of precomputed information in reducing computational difficulty. The complexity class $DTIME(T(n))/a(n)$ consists of problems solvable by a Turing Machine in $T(n)$ steps given access to a size-dependent advice string of length $a(n)$. The advice string is computed once for each input length n and is then applied to every input of that length. Crucially, the advice string is instance-independent: it does not depend on the specific content of the input x , only on its length.

The key results about advice complexity establish that polynomial advice is sufficient to collapse many intractability results: $P/poly$, the class of problems solvable in polynomial time with polynomial advice, contains NP if and only if the polynomial hierarchy collapses (Karp and Lipton, 1980). This means that precomputed information of sufficient quality can in principle make computationally intractable problems tractable, but the quality required for this is itself extremely demanding.

3.3 Distributed Computing and the Byzantine Generals Problem

A distributed system consists of n processors connected by communication channels, each executing a local program and exchanging messages with neighbors. A distributed algorithm specifies, for each processor, the program it executes and the messages it sends in response to received messages and local computation. The communication complexity of a distributed task is the minimum number of bits that must be exchanged across the system for all processors to jointly compute the target function (Kushilevitz, 1997).

The Byzantine Generals problem (Lamport, Shostak and Pease, 1982) studies the conditions under which n processors can reach reliable consensus when t of them are faulty, where a faulty processor may send arbitrary messages, fail to send messages, or actively mislead other processors. The problem formalizes the challenge of coordination in the presence of unreliable or adversarial participants.

The critical results are: (1) Under synchronous protocols (where all messages arrive within a known bounded delay), consensus is achievable if and only if the number of faulty processors satisfies $t < n/3$. This means that up to one-third of the processors can be faulty without preventing consensus. (2) Under asynchronous protocols (where message delays are unbounded), Fischer, Lynch and Paterson (1985) proved that deterministic consensus is impossible even with a single faulty processor. This is the FLP impossibility result, one of the most fundamental negative results in distributed computing theory.

The FLP result is not a practical limitation that better technology can overcome; it is a mathematical impossibility result. In any asynchronous distributed system with even one potentially faulty node, there exists no deterministic protocol that guarantees both safety (no two processors decide differently) and liveness (every non-faulty processor eventually

decides). The choice is between allowing the system to decide incorrectly or allowing it to wait forever.

3.4 Self-Stabilization and Error Correction

A distributed system is self-stabilizing (Dijkstra, 1974) if, starting from any configuration, it eventually reaches and remains in a legitimate configuration without external intervention. Self-stabilization formalizes the notion of a system that can autonomously recover from perturbations, regardless of their origin or magnitude.

The conditions for self-stabilization are demanding. In general, a system can self-stabilize only if its legitimate states form a closed set under the system's transition function, and if the system has sufficient redundancy to detect and correct errors at the rate at which they are introduced. If the error introduction rate exceeds the system's correction capacity, faults accumulate and the system drifts away from legitimate states.

Error-correcting codes (MacWilliams and Sloane, 1977) formalize the relationship between redundancy and error-correction capacity. A code with minimum distance d can detect $d-1$ errors and correct $\lfloor (d-1)/2 \rfloor$ errors. The key insight for institutional analysis is that error-correction capacity is a direct function of structural redundancy: systems with insufficient redundancy relative to their fault exposure rate cannot maintain consistent state.

4. The EPT-EGT-Computation Synthesis

4.1 Formal Mapping: CLI as Circuit Depth

I propose the following formal mapping between the CLI and circuit-theoretic concepts. The proposal is not a proof; it is a theoretical identification that generates testable implications.

Definition 4.1 (Institutional Circuit). Let I be a legal institution. The institutional circuit $C(I)$ is the directed acyclic graph in which: (a) input nodes represent reform proposals entering the institutional processing network; (b) output nodes represent implemented normative changes; (c) intermediate nodes represent institutional actors (courts, professional associations, enforcement agencies, academic institutions) capable of either forwarding a modified signal or generating a blocking signal; and (d) edges represent the formal or informal channels through which normative signals propagate from one actor to another.

Definition 4.2 (Institutional Circuit Depth). The depth $d(I)$ of an institutional circuit is the length of the longest path from any input node to any output node. Equivalently, it is the minimum number of sequential institutional processing steps that any reform signal must traverse before producing a normative output.

Claim 4.1 (CLI as Circuit Depth Proxy). The CLI score of an institution I is a normalized proxy for the circuit depth $d(I)$. Specifically, CLI values in the range $[0, 0.45]$ correspond to circuits in the NC regime (polylogarithmic depth, feasible parallel computation); CLI values in the range $(0.45, 0.70]$ correspond to circuits approaching the NC boundary; and CLI values above 0.70 correspond to circuits with superpoly-logarithmic depth in which sequential bottlenecks make reform computationally intractable under realistic resource constraints.

The four components of the CLI (precedent density, doctrinal elaboration, professional organization, enforcement infrastructure) measure different dimensions of circuit depth.

Precedent density measures the depth of the judicial processing branch: high precedent density means that any reform signal must pass through a long chain of citational dependencies before it can be processed without generating a reversal signal from the precedent network. Doctrinal elaboration measures the depth of the academic processing branch: a highly elaborated doctrine creates a dense network of inferential dependencies that any reform proposal must navigate without triggering refutation signals from within the doctrinal structure. Professional organization measures the coordination depth of the blocking response: highly organized professions can generate blocking signals at every level of the circuit simultaneously, increasing the effective depth by creating simultaneous blocks at multiple layers. Enforcement infrastructure measures the terminal processing depth: even if a reform signal traverses all prior layers, it must still be instantiated through a specific enforcement apparatus, and high infrastructure density means that this final processing stage has its own sequential bottlenecks.

The mathematical structure of CLI as a depth proxy generates a specific prediction about the shape of the reform failure rate curve. If reform failure is a function of circuit depth, and circuit depth is proxied by CLI, then the failure rate function $f(\text{CLI})$ should exhibit three regimes: near-linear in the NC range (0 to 0.45), superlinear in the transition range (0.45 to 0.70), and near-total in the deep-circuit range (above 0.70). This is a more specific prediction than the prior EPT threshold hypothesis, which posited a step function at 0.70.

4.2 IEI as Non-Uniform Reconfiguration Capacity

The IEI is a four-component measure of institutional evolvability: Variation (V), Selection (S), Complexity (C), and Modularity (M). Following the Gould-Dawkins synthesis in the EPT programme, V and M measure structural constraints (what architectures are possible), while S and C measure functional selection pressures (what variants survive within the possible space). The formal mapping to computational complexity is:

V (Variation) maps to the range of circuit architectures available to the institution: the set of distinct circuits $\{C_n\}$ that the system can adopt in response to inputs of different types. High V means the institution can reconfigure its processing structure substantially in response to different types of reform challenge; low V means it applies the same circuit regardless of the reform type.

S (Selection) maps to the differential fitness of alternative circuit architectures: which of the available architectures are selected for by the institutional environment. High S combined with high V creates an evolutionarily dynamic system that can efficiently find and adopt the fittest circuit architecture for each class of inputs.

M (Modularity) maps to the independence of circuit subcircuits: the degree to which different institutional subsystems can be reconfigured independently without affecting each other. High modularity is the formal analog of the NC property: a highly modular institution can process reform signals in parallel across its subsystems, reducing the effective depth of the overall circuit.

C (Complexity) maps to the number of interdependencies between institutional modules: high institutional complexity corresponds to high gate fan-in in the circuit, which increases the depth required to compute any specific output because each gate depends on more prior computations.

The formal implication is that $\text{IEI} = f(V, S, M, C)$ where V and M increase evolvability and C decreases it; S moderates the trade-off between V and M. A high-IEI institution can operate in the non-uniform computation regime: it adopts different processing architectures for different

reform challenges, reducing the effective circuit depth for each specific input class. A low-IEI institution is committed to a uniform architecture regardless of input type: it applies the same deep circuit to every reform proposal, making all reform attempts computationally equivalent in difficulty.

4.3 HBU as Advice-String Preloading

The advice-string interpretation of HBU is the most novel contribution of this synthesis, because it generates a prediction that is not derivable from the evolutionary framework alone: reform windows should coincide with disruptions to professional socialization processes, not merely with political opportunity structures.

In the advice-string formalism, the advice string a_n for inputs of length n is computed once, prior to the arrival of any specific input of length n , and is then supplied to the Turing Machine (or circuit) when processing any such input. The advice is size-dependent: the appropriate advice for a reform proposal of type T is computed during the formation of professional priors about type- T issues, not during the processing of any specific type- T reform proposal.

HBU is the mechanism by which institutional actors acquire their advice strings. During legal education, students observe the reactions of high-authority figures (judges, senior practitioners, prominent academics) to normative propositions. These reactions are convergent within any given institutional environment: students in Argentine labor law programs observe that Article 14bis challenges reliably generate high-intensity negative reactions from virtually all authority figures. This convergence of reactions across authority sources is the analog of the advice string being computed to high precision: the institutional actor has acquired, through reaction observation, a very precise prediction of what the correct response to any type- T input is, before any specific type- T input arrives.

The key implication follows from the precision of the advice string. An advice string that has been computed to high precision effectively reduces the circuit depth required to process subsequent inputs of the corresponding type to near zero: the institution does not need to compute its response from first principles because the response has been preloaded. This is why institutional resistance to reform appears automatic rather than deliberative: the blocking response is the execution of a preloaded advice string, not the outcome of a new computation.

The disruption prediction follows directly. If reform windows require advice strings to be unloaded or corrupted before new normative inputs can be processed without triggering automatic blocking, then reform windows should appear when professional socialization processes are disrupted (curriculum changes, bar exam reforms, forced turnover in judicial appointments, catastrophic professional norm violations that damage the credibility of authority figures). These disruptions corrupt the advice strings that would otherwise generate automatic blocking responses, creating a window in which the full circuit depth must be traversed and in which the reform signal has a non-trivial probability of reaching the output layer before generating a blocking signal from a compromised advice string.

4.4 IHR as Fault-Propagation Cost

The IHR measures the rate at which institutional deformation persists and propagates irreversibly after a perturbation. In distributed computing terms, IHR measures the fault-propagation cost of the institutional circuit: the rate at which a local fault at one node spreads to infect the global state of the system, relative to the system's capacity to detect and correct the fault before it propagates.

The formal connection is through self-stabilization theory and error-correcting codes. A system with IHR close to zero has high error-correction capacity relative to its fault introduction rate: local perturbations are detected and corrected before they propagate, and the system returns autonomously to legitimate states after disturbances. A system with IHR close to one has error-correction capacity that is overwhelmed by the fault introduction rate: local perturbations propagate through the system faster than they can be corrected, generating persistent and expanding deformation.

The Argentina Article 14bis case illustrates this structure. The estimated IHR of 0.87 for this provision (Lerer, 2025c) implies that institutional deformations introduced by reform attempts persist and spread at a rate that far exceeds the system's capacity for autonomous correction. Each failed reform attempt introduces a local deformation (contradictory judicial decisions, inconsistent enforcement, fragmented professional opinion) that propagates through the citation network, the professional training infrastructure, and the enforcement apparatus faster than the system can resolve it. The result is that failed reforms make the next reform attempt harder, not easier: each failed attempt increases circuit depth by adding new layers of contradictory precedent that subsequent reform signals must navigate.

4.5 Zombie Law Equilibria as BFT Threshold Violations

A Zombie Law Equilibrium is the condition in which CLI exceeds 0.85 and institutional fragmentation (the proportion of institutional units holding divergent operative interpretations of the same normative provision) exceeds 0.60 simultaneously. Under these conditions, the legal system continues to process normative inputs formally but cannot produce consistent outputs: different courts apply different standards, enforcement agencies operate under incompatible criteria, professional associations transmit divergent doctrines.

The BFT theorem establishes that under synchronous protocols, consensus is achievable if and only if faulty nodes constitute fewer than $n/3$ of the total. When faulty-node proportion exceeds this threshold, no synchronous protocol can guarantee consensus. In the Zombie Law context, "faulty" institutional nodes are those operating with an internal normative state that diverges from the official normative output of the system. Courts that apply superseded doctrines, enforcement agencies that implement formally repealed standards, academic institutions that transmit jurisprudential positions at odds with current legislation: all of these are faulty nodes in the BFT sense.

The FLP impossibility result extends this to asynchronous systems: in any asynchronous institutional network with even one faulty node, no deterministic protocol can guarantee both consistent normative output and eventual decision. Institutional networks with sufficiently large asynchrony components (because not all actors update their normative states at predictable times) are formally incapable of achieving consensus through ordinary institutional procedures, regardless of the content of the reform and regardless of the political majority behind it.

This characterization adds explanatory content to the prior EPT account. EPT explains why Zombie Law Equilibria are evolutionarily stable: each of the competing normative memes is fit enough within its institutional niche to persist against invasion by rivals. The BFT framework adds the structural reason: the fraction of faulty nodes has exceeded the consensus threshold, and no internally generated protocol can restore consensus. The only resolution mechanism is external protocol change: a constitutional convention that resets the normative baseline for all nodes simultaneously, an international treaty obligation with direct effect that

bypasses the existing circuit, or an administrative regime change that replaces the existing circuit architecture with a new one before the old circuit can generate blocking signals.

5. Empirical Predictions and Test Designs

5.1 Prediction 1: Superlinear Reform Failure Scaling with Circuit Depth

Prediction 1: Reform failure rates as a function of CLI should exhibit three regimes: approximately linear in the range [0, 0.45]; superlinear (approximately quadratic) in the range (0.45, 0.70]; and near-total (above 90%) in the range (0.70, 1.00]. The prior EPT prediction was a step function at 0.70: below 0.70, reform has positive probability; above 0.70, reform fails. The computational prediction is more specific: it predicts the shape of the failure curve, not merely the location of a threshold.

The existing sixty-case international dataset (Lerer, 2025a) provides an initial test. Table 2 reports the observed failure rates by CLI range.

CLI Range	N attempts	N failures	Failure rate
0.00 - 0.45	18	4	22.2%
0.45 - 0.70	22	18	81.8%
0.70 - 1.00	20	19	95.0%

Table 2: Reform failure rates by CLI range, sixty-case international dataset. Source: Lerer (2025a), reanalyzed.

The pattern in Table 2 is consistent with the superlinear scaling prediction. The jump from 22.2% to 81.8% across the transition range is larger than the jump from 81.8% to 95.0% across the deep-circuit range. This is the signature of superlinear rather than step-function scaling: the steepest increase in failure rate occurs in the transition regime, not at the threshold. The dataset is too small for formal statistical tests of the curve shape, but the direction of the evidence is consistent with Prediction 1.

The test that would distinguish Prediction 1 from the prior EPT step-function prediction requires a larger dataset with finer CLI granularity, particularly observations in the 0.45-0.70 range. The current dataset has seventeen cases in this range; a dataset of one hundred or more would allow curve-fitting with sufficient power to distinguish step-function from superlinear shapes.

5.2 Prediction 2: Advice-String Disruption Precedes Reform Windows

Prediction 2: Reform attempts that are preceded (within a five-year window) by disruptions to the professional socialization infrastructure in the relevant doctrinal area should have higher success rates than reform attempts not preceded by such disruptions, controlling for CLI level. The disruptions that the advice-string interpretation predicts to be relevant include: major curriculum changes in law school programs covering the relevant doctrine; reform or abolition of mandatory certification or bar examination requirements in the relevant specialty; forced turnover in judicial appointments covering the relevant courts (through constitutional reform,

term limits, or political discontinuity); and high-profile scandals or institutional failures that damage the credibility of authority figures in the relevant professional community.

This prediction is not derivable from the prior EPT or EGT formulations. EPT predicts that reform windows appear when the fitness of the incumbent normative meme declines relative to the reform meme; EGT predicts that reform windows appear when the payoff landscape shifts sufficiently to make the reform meme an ESS. Neither tradition specifies that the mechanism operating at the professional socialization level is a necessary condition for reform success, as distinct from a contributing factor.

The test requires a dataset of reform attempts paired with data on the professional socialization infrastructure in the five years preceding each attempt. No such dataset currently exists. Constructing one requires: (a) identifying the relevant professional certification requirements and law school curricula for each jurisdiction and doctrinal area in the dataset; (b) coding disruptions to these institutions in the five years preceding each reform attempt; and (c) regressing reform success on disruption indicator, controlling for CLI level. The temporal structure of the prediction (disruption must precede the reform, not accompany it) creates a clean identification design if disruptions are driven by factors independent of the reform agenda.

The 1991 Argentine flexibilization decrees provide a suggestive preliminary test. The decrees were issued in the immediate aftermath of the hyperinflation shock of 1989-1990, which generated: (a) a wave of emergency labor agreements signed under conditions that bypassed ordinary institutional processing; (b) a temporary collapse in the credibility of the professional consensus that had governed labor law interpretation since the 1974 *Procedimiento Laboral*; and (c) significant turnover in the academic leadership of labor law programs, driven by the broader institutional crisis of the Alfonsín administration. The decrees achieved partial success, holding for thirty-six months before the institutional circuit reconstituted its advice strings and generated the blocking signals that eventually rolled them back. This pattern is consistent with Prediction 2: the socialization disruption created a temporary window, but the advice strings were reloaded within three years.

5.3 Prediction 3: Zombie Law Resolution Requires External Protocol Change

Prediction 3: Zombie Law Equilibria ($CLI > 0.85$ and institutional fragmentation > 0.60) should be resolvable only through interventions that are external to the ordinary institutional protocol. Ordinary legislative reform, even with supermajority support, should be insufficient. The intervention types predicted to be effective are: constitutional conventions that reset the normative baseline for all nodes simultaneously; international treaty obligations with direct effect that impose a new normative state on all nodes from outside the circuit; and administrative regime changes that replace the circuit architecture itself before the old circuit can generate blocking signals in response to the new normative input.

The prediction is derived directly from the FLP impossibility result. In an asynchronous distributed system with faulty-node proportion exceeding the BFT threshold, no deterministic protocol can guarantee consensus. A legislative majority is a deterministic protocol: it specifies a decision rule that, given sufficient support, is supposed to produce a specific output. But if the institutional circuit has exceeded the BFT threshold, the legislative decision rule cannot guarantee that the output it produces will be consistent with the state of all institutional nodes: some nodes will continue to apply the old standard, the circuit will remain fragmented, and the Zombie Law will persist alongside the new legislation.

Three cases in the sixty-case dataset are identified as Zombie Law Equilibria on the CLI and fragmentation criteria. In all three, ordinary legislative reform failed to resolve the fragmentation. Resolution, when it occurred, came from external protocol changes: in the Argentine CGT injunction case (Lerer, 2026b), a mandatory coordination protocol between the executive and the judicial review function, which bypassed the ordinary appellate circuit; in the Chilean labor fragmentation case of 2008-2011, a constitutional amendment that redefined the interpretive mandate of the labor courts, effectively replacing the existing circuit architecture; and in the Spanish banking resolution case of 2012-2014, European Central Bank intervention that imposed normative uniformity from outside the domestic circuit.

5.4 Platform 3: ABM Extension

The 816-agent computational model (Lerer, 2026c) provides a third testing platform for the synthesis. The current model implements HBU as a learning rule and CLI as an environmental parameter; it produces distributions of reform outcomes across parameter regimes. The synthesis generates two specific extensions for this model.

Extension 1: Advice-string saturation dynamics. The current model treats advice strings as static: agents arrive with preconditioned priors that remain fixed across simulation time. The advice-string interpretation predicts that the rate of advice-string reconditioning in response to authority-reaction disruptions should be a key parameter governing reform dynamics. Adding this parameter and varying it should produce phase transitions in reform success rates at critical reconditioning thresholds, which would be consistent with Prediction 2.

Extension 2: BFT threshold identification. The current model does not identify the threshold at which Zombie Law Equilibria form. Adding a fragmentation tracker to the model (measuring the proportion of agents operating with divergent normative states at each time step) and identifying the CLI-fragmentation combinations that produce irreversible fragmentation would allow in silico testing of the BFT threshold prediction. The formal prediction is that irreversible fragmentation should appear when the product of CLI and fragmentation rate exceeds the BFT threshold, which the model can evaluate directly.

6. Cross-Country Evidence: The Four-Case Comparison

The four-country comparison provides an initial coherence check for the synthesis. The sample is too small for causal inference; the purpose is to verify that the computational reinterpretation is consistent with the observed pattern and adds explanatory content relative to prior accounts.

6.1 Argentina: Deep-Circuit Regime with Advice-String Saturation

Argentina's labor regime exhibits $CLI=0.87$, placing it firmly in the deep-circuit regime. The circuit depth is contributed roughly equally by the four CLI components: 187 specialized labor tribunals (precedent density), mandatory labor law curricula in all 65 ABA-accredited law schools with bar examination requirements (doctrinal elaboration), 185 bar associations with demonstrated rapid-response coordination capacity (professional organization), and 50,000+ labor law specialists with certification requirements tied to the existing doctrinal structure (enforcement infrastructure).

The advice-string interpretation adds a temporal dimension to this characterization. The circuit did not reach its current depth instantaneously; it accumulated over seventy years of

socialization cycles since the 1949 constitutionalization of labor rights. Each socialization cycle reloaded the advice strings of a new cohort of legal professionals, adding them to the blocking architecture. The result is a circuit in which the advice strings are not merely present but have been replicated across multiple generations, creating a depth of saturation that makes disruption-based reform windows extremely narrow and short-lived.

The twenty-three reform failures span a wide range of political conditions, but the computational framework predicts that political conditions are second-order relative to circuit depth. A system with $CLI=0.87$ is predicted to have a 95% failure rate regardless of the political configuration behind the reform attempt, because the circuit depth itself generates sequential blocking requirements that no reform proposal can traverse before generating a blocking signal. The single partial success (1991-1994) is consistent with the advice-string disruption prediction: it occurred in the immediate aftermath of a hyperinflation shock that temporarily disrupted professional socialization, and it ended when the socialization process reconstituted the saturated advice strings.

6.2 Brazil: Shallow Circuit with Self-Stabilizing Capacity

Brazil's pre-2017 CLI of 0.40 corresponds to the transition regime approaching the NC boundary. The circuit was not deep: precedent density was lower (15 specialized labor tribunals against 187 in Argentina), doctrinal elaboration was less centralized (the CLT's regulatory structure is statutory rather than constitutional, creating weaker inferential dependencies), professional organization was less coordinated (union density comparable but collective bargaining fragmented across sectors), and enforcement infrastructure was less specialized (no bar examination requirement keyed to specific doctrinal commitments).

The 2017 reform succeeded without triggering BFT threshold violations because the advice-string density was insufficient to generate automatic blocking responses across all nodes simultaneously. Individual courts and individual professional associations resisted, but the resistance was not coordinated in a way that traversed the circuit in parallel, and the reform signal was able to reach the output layer before the blocking signals from individual nodes accumulated to circuit-wide blocking force.

The post-2017 evolution is interesting for the synthesis. The reform introduced new doctrinal content that has been transmitted through subsequent socialization cycles, incrementally increasing the CLI. If the computational interpretation is correct, Brazil's labor regime should show increasing reform resistance over the coming decade as the new doctrinal structure builds its circuit depth.

6.3 Chile: Non-Uniform Computation in Practice

Chile's CLI of 0.24 corresponds to the NC regime: low depth, high modularity, capacity for efficient parallel processing of reform inputs. The three successful reforms in fifteen years are consistent with the non-uniform computation interpretation: each reform adopted a different institutional architecture rather than applying a uniform procedure.

The 1990 reform (post-Pinochet democratic transition) operated through constitutional mandate, effectively replacing the circuit architecture before the existing circuit could generate blocking signals. The 2001 reform operated through negotiated collective bargaining restructuring, exploiting the modular independence of different sectors of the labor market to implement reform in parallel across sectors rather than sequentially through a central circuit. The 2015 reform operated through the Constitutional Tribunal's interpretive mandate,

effectively inserting a new processing layer at the apex of the circuit that reconfigured the processing architecture before the reform signal reached the lower layers.

Each reform exploited the non-uniform reconfiguration capacity of Chile's institutional system: the ability to choose the circuit architecture appropriate for the specific type of reform input. This is the IEI advantage in computational terms: high evolvability means the system can find the efficient processing architecture for each reform challenge, rather than being forced to run all reform signals through the same deep circuit.

6.4 Spain: Threshold Proximity and External Protocol Sensitivity

Spain's CLI of 0.37 occupies the transition zone between the NC regime and the deep-circuit regime. Prior EPT work identified Spain as a potential refutation of the threshold hypothesis: CLI of 0.37 is below the 0.70 threshold but above the 0.45 NC boundary, and both the 1994 and 2012 reforms succeeded, apparently confirming the prediction that sub-threshold CLI allows reform.

The computational reinterpretation accommodates Spain more precisely. A circuit in the transition zone (0.45-0.70) has positive but declining probability of reform success under ordinary political conditions; external protocol changes should have disproportionate effectiveness in this range because they can tip the balance without needing to fully replace the circuit architecture.

The 1994 reform succeeded under ordinary political conditions (a Socialist government with parliamentary majority) because Spain's circuit, while not in the NC regime, had not yet built the sequential bottlenecks of the deep-circuit regime. The 2012 reform required EU-level external pressure (conditionality linked to the European Stability Mechanism program) to succeed: by 2012, the circuit had deepened sufficiently that ordinary political conditions were no longer sufficient, but external protocol change could provide the supplementary pushing force necessary to traverse the circuit. This is precisely what the BFT-threshold framework predicts for circuits near the consensus threshold: ordinary protocols are insufficient, but external signals (analogous to synchrony assumptions in distributed computing) can restore feasibility.

7. Toward a Formal Mapping: Preliminary Axioms

The synthesis proposed in this paper rests on structural analogies between institutional metrics and computational concepts. This section makes the mapping more precise by specifying, in axiomatic form, the conditions under which the analogy holds formally rather than merely heuristically.

7.1 The Institutional Circuit Axioms

The institutional circuit representation requires the following conditions to hold for any institution I:

Axiom 1 (Finite Node Set). The set of institutional actors capable of processing normative signals in the domain of I is finite and identifiable. This axiom is satisfied in practice because legal systems have countable institutional structures (courts, agencies, associations) with defined membership criteria.

Axiom 2 (Acyclicity). The processing relationships among institutional actors, when restricted to a specific reform proposal, form a directed acyclic graph. This axiom is the most substantively demanding. Institutional processing relationships are frequently cyclic in general (courts cite each other, agencies consult academic doctrine, academics analyze court decisions), but for any specific reform proposal, the processing sequence has an initial input (the reform proposal) and a terminal output (implementation or blocking), with finite intermediate processing steps. The restriction to a specific reform proposal is what induces acyclicity.

Axiom 3 (Signal Monotonicity). A normative signal either passes through a processing node (possibly transformed) or generates a blocking signal. It does not increase in force after a blocking signal has been generated. This axiom formalizes the intuition that blocking generates blocking: a constitutional challenge filed at one level does not enhance the probability of implementation at other levels.

Axiom 4 (Depth Computability). Given a specification of the institutional actors and their processing relationships, the depth of the resulting circuit is computable. This axiom is satisfied by construction, since circuit depth is defined as the length of the longest path in the DAG, which is computable in linear time for any finite DAG.

7.2 The Advice-String Axioms

The advice-string interpretation of HBU requires:

Axiom 5 (Size-Dependence). The content of the advice string loaded during professional socialization depends on the type and scale of the normative area (the "input size") but not on the content of any specific reform proposal within that area (the "specific input instance").

Axiom 6 (Pre-Instance Loading). The advice string is loaded before any specific reform proposal in the relevant area arrives at the institutional circuit for processing.

Axiom 7 (Precision Monotonicity). The precision of the advice string (measured as the inverse variance of the normative prior it generates) is a non-decreasing function of the number of socialization cycles in which that string has been replicated. More socialization cycles produce more precise advice strings, which generate more certain blocking responses.

These axioms are empirically testable: Axiom 5 predicts that the blocking responses to reform proposals within a given doctrinal area should be type-invariant (different reforms in the same area should generate similar blocking patterns); Axiom 6 predicts that blocking responses should appear before the specific content of the reform proposal has been evaluated on its merits; and Axiom 7 predicts that doctrinal areas with longer histories of socialization should show lower variance in professional blocking responses.

7.3 Limits of the Formalization

Three limitations of the formal mapping require acknowledgment.

First, the mapping between CLI component weights and circuit depth contributions is proposed but not formally derived. The claim that CLI=0.87 corresponds to a specific circuit depth value requires either a formal proof that CLI satisfies the axioms of a depth measure, or an empirical calibration that maps CLI scores to observed blocking patterns. Neither has been done in this paper; both are research priorities.

Second, the BFT analogy requires counting faulty nodes, which requires a definition of "faulty" in institutional terms. The current operationalization (nodes with divergent operative interpretations) is a composite index rather than a node count, and the aggregation method may not preserve the formal properties of node-counting that the BFT theorems require.

Third, the FLP impossibility result applies to deterministic consensus protocols in asynchronous systems. Institutional systems have probabilistic elements (discretionary judicial decisions, administrative interpretive leeway) that may make them formally probabilistic rather than deterministic systems. Randomized consensus protocols (Ben-Or, 1983) are feasible in asynchronous systems even with faulty nodes; it is possible that institutional systems are better modeled as randomized than as deterministic, which would require a different impossibility framework.

8. A Research Agenda for the Synthesis

8.1 Immediate Priorities

The most immediate empirical priority is the construction of the socialization disruption dataset for testing Prediction 2. The dataset requires coding professional socialization infrastructure changes for the sixty cases in the existing international dataset, plus a minimum of forty additional cases to achieve adequate statistical power for the curve-fitting required to test Prediction 1 rigorously. Preliminary conversations with comparative labor law scholars suggest that the data for Latin American cases (Argentina, Brazil, Chile, Colombia, Mexico, Peru) can be reconstructed from bar association records and law school curriculum archives. European cases require access to professional certification records that may require institutional partnerships.

The second immediate priority is the ABM extension specified in Section 5.4. The 816-agent model is operational and the two extensions (advice-string reconditioning dynamics and BFT threshold tracking) can be implemented within the existing architecture. Preliminary simulation results from the advice-string extension are in preparation.

The third immediate priority is the formalization of the CLI-to-circuit-depth mapping, which requires either the formal proof approach or the empirical calibration approach described in Section 7.3. The empirical calibration approach is more tractable in the short term: it requires selecting a subset of the sixty cases for detailed process-tracing, coding the specific institutional blocking sequences, counting the sequential processing layers traversed before blocking or implementation, and regressing the resulting depth counts on CLI scores. This would provide an empirical calibration of the CLI-depth relationship that could validate or revise the theoretical mapping.

8.2 Medium-Term Directions

The Price Equation (Price, 1970) offers a formal bridge between the evolutionary and computational components of the synthesis that has not been fully exploited in this paper. The Price Equation decomposes the change in any population-level trait across a generation into a selection component and a transmission component: $\Delta(z) = \text{Cov}(w, z)/\bar{w} + E(w * \Delta(z))/\bar{w}$. Applied to institutional analysis, the selection component measures the differential fitness of competing normative strategies; the transmission component measures how the content of strategies changes during replication. The computational interpretation is

that the selection component corresponds to the circuit-level fitness differential (circuits with lower depth are selected for in environments with high reform pressure), while the transmission component corresponds to the advice-string fidelity (how accurately the preloaded responses are transmitted across socialization cycles).

The TRIBE v2 neuroscience experiments (arms 1 and 2) offer a potential empirical window into the advice-string precision claim at the level of individual cognition. If HBU produces advice strings through a reaction-observation learning mechanism, and if advice-string precision increases with socialization intensity, then neuroscientific measures of normative processing (ERP components associated with normative violation detection) should show systematic differences between legal professionals at different career stages and in different doctrinal specializations. The computational prediction is specific: the latency and amplitude of normative violation signals should decrease with socialization depth (faster and stronger responses reflecting higher-precision advice strings), and the response to out-of-domain violations should be weaker and more variable than the response to in-domain violations (consistent with the size-dependence axiom).

8.3 Long-Term Agenda

The long-term agenda of the synthesis is the development of a general computational theory of institutional evolution that can address the macro-historical questions Hamilton et al. (2026) identify as central: why do institutional hierarchies emerge, how do they restructure with population scale, and why do external memory systems (legal codes, constitutional texts, jurisprudential traditions) take the forms they do.

The EPT-EGT programme has addressed these questions from the evolutionary side; the computational tradition has addressed them from the information-processing side. The synthesis suggests that the answers are aspects of the same structural phenomenon: institutional hierarchies emerge because they reduce circuit depth relative to non-hierarchical alternatives (the NC computation advantage); they restructure with scale because the optimal non-uniform circuit architecture changes as input complexity grows; and external memory systems take their specific forms because they are the advice strings of institutional circuits, and the format of advice strings is constrained by the precision requirements of the blocking mechanisms they serve.

9. Limitations

Beyond the formal mapping limitations discussed in Section 7.3, three substantive limitations warrant explicit acknowledgment.

First, the synthesis proposed here is theory-forward: it generates predictions that follow from the formal integration, but the formal integration itself rests on analogical reasoning whose limits are not fully specified. The claim that CLI is a proxy for circuit depth is plausible and generates coherent predictions, but it requires formal validation that has not yet been completed. Papers in this programme that have generated predictions without completing their empirical validation before publication have required corrections (the Lateral Silencing Exaptation paper, v2, is an example); the same risk applies here, and readers should treat the synthesis as a research programme specification rather than a validated theory.

Second, the paper does not address the normative dimension of the phenomena it describes. Computational intractability is a structural fact; it does not imply that resistant institutions are good or bad, or that reform is desirable or undesirable. The programme's empirical work has consistently abstracted from normative evaluation to focus on structural analysis. This abstraction is methodologically defensible but should not be read as normative neutrality: the structural analysis is a precondition for effective normative intervention, not a substitute for it.

Third, the four-case comparison in Section 6 is not a causal identification design. The paper does not claim to have established that CLI or circuit depth cause reform failure; it claims that the computational framework is consistent with the observed pattern and generates more specific predictions than the prior EPT formulation. Causal identification requires the research designs specified in Section 5, which have not yet been executed.

10. Conclusion

The convergence between the EPT-EGT research programme and the computational social science tradition of Hamilton et al. (2026) is an instance of independent derivation: two research communities, working from different starting points, using different methods, and asking different primary questions, have arrived at structurally similar claims about the same phenomena. The EPT-EGT programme describes institutional persistence through the lens of evolutionary dynamics; the computational tradition describes it through the lens of information-processing resource constraints. The synthesis of these two perspectives is more powerful than either alone because it supplies what each lacks.

The synthesis generates three falsifiable predictions that extend beyond the prior EPT formulation. Reform failure rates should exhibit superlinear (not step-function) scaling with CLI. Reform windows should coincide with professional socialization disruptions, not merely with political opportunity structures. Zombie Law Equilibria should require external protocol changes for resolution, not legislative majorities. Each prediction specifies the empirical test that would refute it.

More fundamentally, the synthesis suggests that the question "why is this institution resistant to reform?" has both an evolutionary answer (the incumbent normative meme is evolutionarily stable) and a computational answer (the circuit depth of the institution exceeds the NC threshold for computationally feasible reform). These two answers are not alternatives; they are descriptions of the same structural reality at different levels of abstraction. The evolutionary description explains why the circuit has the depth it has; the computational description explains what constraints that depth imposes on feasible reform trajectories.

The immediate implication for reform strategy is that political majority is a necessary but not sufficient condition for reform success in high-CLI institutions. The sufficient condition involves either reducing circuit depth before the reform attempt (by disrupting the socialization processes that maintain advice-string saturation) or changing the processing protocol itself (by intervening at the architectural level rather than at the input level). This is not a counsel of despair; it is a specification of the problem that effective reform strategy must address.

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